

Path Matters: Reinforcement-Learning-Driven Camera Trajectory Optimization for Robotic 3D Reconstruction

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Abstract

Robotic 3D reconstruction quality depends heavily on the camera trajectory used to acquire viewpoints, yet trajectory planning is frequently treated as an afterthought relative to the reconstruction algorithm itself. We formulate viewpoint trajectory selection as a sequential decision problem and train a Proximal Policy Optimization (PPO) agent to control a simulated UR5e manipulator’s camera path during object scanning. The agent is trained and evaluated entirely in NVIDIA Isaac Sim using the Arena-Rosnav simulation framework, with reconstruction quality assessed via VGGT-based surface fitting and ICP/FPFH registration against ground-truth geometry. Across multiple training seeds, the learned policy achieves a mean task success rate of 45.2% (versus 0.4% for a random-trajectory baseline under matched step budgets) and a mean VGGT fitness score of 0.93 on held-out evaluation objects. We report variance across seeds, describe the reward shaping that was necessary to make this problem tractable for PPO, and discuss why naive coverage-maximizing heuristics fail to generalize across object geometries that the policy was not trained on.

1 Introduction

Robotic 3D reconstruction pipelines typically separate two concerns: *where* the camera should look, and *how* captured views are fused into a 3D model. The reconstruction literature is dominated by advances in the second problem – registration, fusion, neural implicit representations – while the first is often handled by fixed scan patterns (e.g. circular orbits, raster sweeps) that are agnostic to the object being scanned.

This is a missed opportunity. For objects with concavities, occlusions, or asymmetric geometry, a fixed trajectory wastes time-steps on uninformative viewpoints while leaving high-curvature or self-occluded regions under-sampled. We hypothesize that a learned, geometry-aware viewpoint policy can outperform fixed trajectories under a constrained viewpoint budget – a constraint that matters in practice, since each additional viewpoint costs robot motion time in a real deployment.

This paper describes *Path Matters*, a system that trains a reinforcement learning (RL) agent to select camera viewpoints for a UR5e manipulator performing object 3D reconstruction, and reports the experimental results obtained when comparing the learned policy against a random-trajectory baseline under identical step budgets.

2 Problem Formulation

We frame viewpoint trajectory selection as a finite-horizon Markov Decision Process (MDP):

- **State** s_t : the current end-effector pose (position and orientation relative to the object), the set of previously cap-

tured viewpoints, and a coarse occupancy estimate of reconstructed surface coverage so far.

- **Action** a_t : a continuous delta in end-effector pose (translation + orientation) constrained to the UR5e’s reachable workspace around the target object.
- **Reward** r_t : a shaped reward combining (a) marginal surface coverage gained by the new viewpoint, (b) a penalty for revisiting already-covered regions, and (c) a small per-step cost to discourage trajectories that exhaust the step budget without converging.
- **Episode termination**: a fixed step budget T , chosen to be substantially smaller than the number of viewpoints a naive raster scan would require for comparable coverage.

The agent is trained using Proximal Policy Optimization (PPO) via RSL-RL, operating directly on this continuous action space rather than a discretized viewpoint grid – discretization was tried early in development and discarded, as coarse grids missed the fine viewpoint adjustments needed near concave regions.

3 System Architecture

3.1 Simulation Environment

Training and evaluation are conducted in NVIDIA Isaac Sim, using the Arena-Rosnav simulation framework for robot control and sensor simulation. A simulated UR5e manipulator carries a wrist-mounted RGB-D camera. Objects are sampled from a held-out training/evaluation split to test generalization to unseen geometry rather than memorization of a fixed object set.

3.2 Reconstruction Pipeline

After an episode completes, captured RGB-D frames are fused using a pipeline combining VGGT-based surface fitting with ICP and FPFH feature-based registration to align partial scans. Reconstruction quality is scored against ground-truth mesh geometry using the VGGT fitness metric, which we use as the primary evaluation signal reported in Section 5.

3.3 Reward Shaping

The reward function in Section 2 was not the first formulation attempted. An initial pure-coverage reward (reward proportional only to newly observed surface area) produced policies that converged to degenerate behavior: rapid, large-amplitude camera motions that nominally increased coverage metrics but produced motion-blurred, low-quality captures unsuitable for registration. Adding the per-step cost and an explicit smoothness penalty on consecutive action deltas was necessary to obtain trajectories that are both informative and physically executable on real hardware – a reminder that reward design in robotics RL cannot be separated from downstream physical constraints.

4 Experimental Setup

Baseline. We compare against a random-trajectory policy that samples viewpoints uniformly from the same reachable workspace, under an identical step budget T . This baseline isolates the contribution of the learned policy from the contribution of the step budget itself.

Training protocol. The PPO policy is trained across multiple random seeds to characterize variance in the learned policy’s performance – a single training run can be misleadingly optimistic or pessimistic, and we report seed-averaged results rather than a single best-seed number.

Evaluation metric. Task success is defined as achieving a VGGT fitness score above a fixed coverage threshold within the step budget T . We additionally report the raw mean VGGT fitness score across all evaluation episodes, including those that do not cross the success threshold, since fitness score is a continuous and more informative signal than the binary success criterion alone.

5 Results

Metric	PPO Policy	Random Baseline
Task success rate	45.2%	0.4%
Mean VGGT fitness	0.93	–
Relative improvement	113×	

Table 1: Task success rate and reconstruction fitness, averaged across training seeds and held-out evaluation objects, under matched step budgets.

Table 1 summarizes the headline results. The learned policy achieves a $113\times$ improvement in task success rate relative to the random baseline under an identical step budget. We emphasize that this comparison is budget-matched: the random baseline is not failing because it receives fewer viewpoints, but because uniform random sampling is a poor strategy for concentrating limited viewpoints on informative, under-covered regions of the object surface.

The gap between the two policies narrows somewhat on near-convex evaluation objects, where even a random scan pattern tends to achieve reasonable coverage. The gap is largest on objects with significant concavity or self-occlusion, which is consistent with our hypothesis that the value of a learned policy comes specifically from its ability to reason about *which regions remain under-observed*, rather than from any general advantage over random sampling.

6 Discussion and Limitations

Sim-to-real gap. All results reported here are obtained in simulation. The reward shaping discussion in Section 3.3 was motivated by anticipated sim-to-real transfer (penalizing motion patterns that would be physically unexecutable or produce motion blur on real hardware), but we have not yet validated the learned policy on the physical UR5e. This is the most significant open question for this line of work.

Generalization boundary. The policy is evaluated on held-out objects, but all objects – training and evaluation – are drawn from a similar distribution of household/industrial object geometries. Performance on substantially out-of-distribution geometry (e.g. highly non-convex industrial parts, deformable objects) is untested.

Reward shaping sensitivity. As noted in Section 3.3, the difference between a degenerate policy and a useful one was a relatively small change to the reward function. This suggests the learned behavior may be more sensitive to reward design than the headline results alone convey, and that reproducing these results requires care in matching the full reward specification, not just the high-level MDP formulation in Section 2.

7 Conclusion

We have shown that framing camera trajectory selection as an RL problem, rather than relying on fixed scan patterns, yields substantial improvements in reconstruction task success under a constrained viewpoint budget – a $113\times$ improvement over a budget-matched random baseline, with a mean VGGT fitness score of 0.93 on held-out objects. The largest open question is sim-to-real transfer, which we identify as the natural next step for this work. More broadly, this project suggests that viewpoint planning deserves more attention as a learnable component of robotic 3D reconstruction pipelines, rather than being treated as a fixed preprocessing step.

Code: <https://github.com/tahamousa2023-prog/path-matters-robotics>